JOB-READY IN 30 DAYS

Machine Learning & AI

Intensive Learning Path

30-day, 10-hours-a-day program · 300 study hours

Target: Applied ML / Generalist role

Built from your three books

Practical Statistics for Data Scientists

Machine Learning with PyTorch and Scikit-Learn

Python Machine Learning by Example

Prepared for Dan · July 2026

Contents

How to use this plan

The goal. Over the next 30 days you will move from "some ML basics" to being able to walk into an Applied ML / Generalist interview and speak fluently about statistics, classical machine learning, and modern deep learning — backed by a portfolio of eight projects you built yourself. This document is your daily operating manual. Each day tells you exactly what to read, what to code, and what to produce.

The three-book strategy. Your books are not read cover-to-cover in isolation. They are braided together by theme so that each concept is hit three ways on the days it matters:

- **Practical Statistics for Data Scientists** — the **why**. Statistical foundations, sampling, experiments, and the math intuition behind models. This is what separates people who run scikit-learn from people who understand it.
- **Machine Learning with PyTorch and Scikit-Learn** — the **how**. Your core technical spine: classical ML in scikit-learn, then deep learning in PyTorch, built up rigorously from first principles.
- **Python Machine Learning by Example** — the **do**. Real, end-to-end projects (recommenders, ad-click prediction, stock prediction, image and text models) that turn theory into a portfolio.

How to read the schedule. Each week is a table: the day, its theme, the exact chapters to read and code along with, and the concrete thing you should have produced by the end of the day.

"PS" = Practical Statistics
"ML" = ML with PyTorch & Scikit-Learn,
"BE" = Python ML By Example.

Non-negotiables for success:

(1) Type every code example yourself — never copy-paste.
(2) Keep a daily learning journal in your Git repo.
(3) Finish each day by explaining the day's key idea out loud in plain English.
(4) Ship every project to GitHub, however rough. Consistency over perfection.

Your daily rhythm (10 hours)

Ten focused hours is a lot; structure protects you from drift and burnout. Use this four-block template every day. Adjust the clock to your life, but keep the proportions and the order — read, then build, then apply, then consolidate.

Block	Time	What you do
Theory & reading	2.5 hrs	Read the day's chapter(s). Take structured notes in your journal: definitions, the intuition, and one question you still have.
Guided build	3.0 hrs	Code along with the book's examples in a notebook. Run every snippet. Break it on purpose, then fix it — that is where learning happens.
Applied / project	2.5 hrs	Apply the day's technique to your own dataset, or push the week's project forward. Aim for one working artifact per day.
Review & interview prep	1.5 hrs	Flashcards, explain-aloud, 1–2 coding/ML interview questions, and log the day. Update your progress tracker.

That is 9.5 hours of work plus a 0.5-hour daily planning-and-reflection buffer. **Take real breaks** between blocks — a rested brain retains far more.

Day 0: before you start (evening prep)

Spend an hour the night before Day 1 so you never lose study time to setup. Get these in place:

1. **Environment:** install Miniconda; create an env (Python 3.11) with numpy, pandas, matplotlib, seaborn, scikit-learn, jupyterlab, and PyTorch (CPU is fine; use Google Colab for GPU-heavy days in Weeks 3–4).
2. **Version control:** create a GitHub repo named ml-30-days. Inside it make a /notebooks, /projects, and /journal folder. Commit daily.
3. **Datasets:** make a free Kaggle account (data + a place to publish notebooks). Bookmark the UCI ML Repository.
4. **Journal:** create journal/README.md with a one-line-per-day template: what I learned, what confused me, tomorrow's goal.
5. **Accounts:** Hugging Face (for Week 4 transformers) and, optionally, a Weights & Biases account for experiment tracking.

Now the 30 days.

Week 1 — Foundations: environment, statistics & first models

Focus: Build rock-solid statistical intuition and implement your first classifiers from scratch. This week is the foundation everything else stands on — do not rush it. By Friday you will understand sampling, significance, and how a linear classifier actually learns.

Day	Theme & focus	Read + code (book · chapter)	Deliverable / practice
1	**Orientation & the ML landscape**	ML Ch1 (ML types, model-building pipeline) · BE Ch1 (ML tasks, data & tools)	Working conda env + initialized GitHub repo; journal README; “hello ML” notebook running end to end
2	**Exploratory Data Analysis**	PS Ch1 (location & variability estimates, distributions, correlation, multivariable EDA)	EDA notebook on a real Kaggle CSV: summary stats, boxplots, histograms, correlation heatmap
3	**Sampling & distributions**	PS Ch2 (CLT, standard error, the bootstrap, confidence intervals, key distributions)	Notebook: bootstrap confidence intervals + a sampling simulation; a short written CLT explainer
4	**Experiments & significance**	PS Ch3 (A/B testing, hypothesis tests, p-values, t-tests, ANOVA, chi-square, power)	Simulate an A/B test + a permutation test; flashcards on p-values and Type I/II errors
5	**ML from first principles**	ML Ch2 (perceptron & Adaline from scratch, gradient descent)	Implement perceptron + Adaline in pure NumPy; plot the decision boundary and the cost curve

Day	Theme & focus	Read + code (book · chapter)	Deliverable / practice
6	**A tour of classifiers**	ML Ch3 (logistic reg, SVM, KNN, trees in scikit-learn) · PS Ch5 (Naive Bayes, discriminant analysis, ROC/AUC)	Train & compare 5 classifiers on one dataset; build a confusion-matrix + ROC evaluation notebook
7	**Project + week review**	BE Ch2 (Naive Bayes movie recommender)	PROJECT 1: movie recommender shipped to GitHub; Week-1 self-quiz (30 questions); journal recap

Milestone: you can explain bias, variance, sampling distributions, and a p-value without notes, and you have implemented a classifier from scratch.

Week 2 — Classical ML mastery

Focus: The bread and butter of applied ML jobs — preprocessing, dimensionality reduction, disciplined evaluation, and the tree/ensemble family that wins most tabular problems. You will build four portfolio projects this week.

Day	Theme & focus	Read + code (book · chapter)	Deliverable / practice
8	**Data preprocessing**	ML Ch4 (missing data, categorical encoding, scaling, feature selection, L1/L2 regularization)	A reusable sklearn preprocessing pipeline (ColumnTransformer + Pipeline) you can drop into any project
9	**Dimensionality reduction**	ML Ch5 (PCA, LDA, kernel PCA) · PS Ch7 (PCA section)	Notebook comparing PCA vs LDA vs t-SNE visualizations on one dataset; written intuition for each
10	**Model evaluation & tuning**	ML Ch6 (cross-validation, pipelines, grid/random search, learning & validation curves, metrics)	A full CV + GridSearchCV tuning workflow; a nested-CV example; a metrics cheat-sheet
11	**Trees, forests & boosting**	PS Ch6 (KNN, trees, bagging, random forest, XGBoost) · BE Ch3 (tree-based ad click-through)	PROJECT 2a: ad click-through — decision tree → random forest → XGBoost; feature-importance analysis
12	**Ensembles & scalable models**	ML Ch7 (voting, bagging, boosting) · BE Ch4 (logistic regression, SGD, online learning, big data)	PROJECT 2b: scalable logistic regression with SGD; compare against your ensembles from Day 11
13	**Regression deep dive**	ML Ch9 (linear, polynomial, tree regression) · PS Ch4 (regression, diagnostics, splines) · BE Ch5 (stock prices)	PROJECT 3: stock-price regression with residual diagnostics + regularization comparison
14	**SVM, clustering & review**	BE Ch9 (SVM face recognition) · ML Ch10 (k-means, hierarchical, DBSCAN) · PS Ch7 (clustering)	PROJECT 4: face recognition (SVM) + a clustering notebook; Week-2 review & interview drills

Milestone: you can take a raw tabular dataset to a tuned, honestly-evaluated model — and defend every choice. Four projects now live on GitHub.

Week 3 — Deep learning

Focus: Neural networks from the ground up, then fluency in PyTorch, then the two workhorse architectures — CNNs for images and RNNs for sequences. Use Colab/GPU on the heavier training days.

Day	Theme & focus	Read + code (book · chapter)	Deliverable / practice
15	**Neural nets from scratch**	ML Ch11 (multilayer NN + backprop from scratch) · BE Ch6 (activation, feedforward, backprop)	Implement an MLP from scratch in NumPy and train it on an MNIST subset; log the loss curve
16	**PyTorch essentials**	ML Ch12 (tensors, autograd basics, GPUs, DataLoaders, building NNs in PyTorch)	Reimplement the MLP in PyTorch with a clean training loop + DataLoader; match your scratch accuracy
17	**PyTorch mechanics**	ML Ch13 (autograd, nn.Module, computation graphs, PyTorch Lightning)	Refactor into an object-oriented model + a Lightning version; save/load checkpoints
18	**Convolutional networks**	ML Ch14 (convolution, pooling, CNN architecture & implementation)	Build and train a CNN image classifier; visualize learned feature maps
19	**CNN project**	BE Ch11 (clothing images, data augmentation, transfer learning)	PROJECT 5: Fashion-MNIST CNN + transfer learning; beat a target accuracy and document ablations
20	**Sequences & RNNs**	ML Ch15 (RNN, LSTM, text generation) · BE Ch12 (sentiment, forecasting, text generation)	PROJECT 6: IMDb sentiment RNN + a character-level text generator
21	**NLP, embeddings & review**	ML Ch8 (sentiment / bag-of-words) · BE Ch7 (text analysis, embeddings, t-SNE) · BE Ch8 (topic modeling)	Newsgroups text-analysis + topic-model notebook; Week-3 review & deep-learning interview drills

Milestone: you can design, train, debug, and evaluate a neural network in PyTorch for both image and text data — and explain backprop from memory.

Week 4 — Modern AI, capstone & interview readiness

Focus: Transformers and generative/multimodal models, then a full-week pivot into a portfolio capstone and hard interview preparation. This is the week that makes you hireable, not just knowledgeable.

Day	Theme & focus	Read + code (book · chapter)	Deliverable / practice
22	**Transformers & attention**	ML Ch16 (attention, the transformer, BERT, GPT) · BE Ch13 (BERT & GPT projects)	Fine-tune/apply BERT for sentiment; a GPT text-generation notebook using Hugging Face
23	**Generative & multimodal**	ML Ch17 (autoencoders, GANs) · BE Ch14 (CLIP image search, zero-shot classification)	PROJECT 7: a GAN on MNIST + a CLIP image-search demo
24	**Graph NNs & reinforcement learning**	ML Ch18 (graph neural networks) · ML Ch19 + BE Ch15 (Q-learning, deep Q, environments)	Q-learning agent on FrozenLake/Taxi; skim & run the GNN molecular demo
25	**Best practices + capstone kickoff**	BE Ch10 (21 ML best practices) · deployment/MLOps overview	Choose your CAPSTONE; write a one-page spec (problem,

Day	Theme & focus	Read + code (book · chapter)	Deliverable / practice
			data, metric, plan); scaffold the repo
26	**Capstone build I**	Apply the full pipeline you have learned	End-to-end: data → EDA → clean baseline model → first honest evaluation
27	**Capstone build II**	Modeling + tuning + (optional) a deep-learning component	Iterate the model, tune rigorously, add a DL component if it fits; error analysis
28	**Capstone finalize**	Packaging & communication	README, results write-up, a simple demo (notebook, Streamlit, or small API); push to GitHub
29	**Interview intensive**	Targeted review across all three books	ML theory Q&A, coding drills, an ML system-design case; polish resume + portfolio
30	**Mock interview & launch**	Consolidation & delivery	Timed mock interview; fix weak spots; capstone presentation; a concrete job-application plan

Milestone: a polished capstone plus seven supporting projects, and the ability to talk through any of them under interview pressure.

Your project portfolio

By Day 30 you will have eight projects on GitHub. Employers hire on evidence, not certificates — each of these is proof you can do the work. Keep every repo clean: a clear README, a requirements file, and a short results section with at least one chart.

#	Project	What it demonstrates
1	Movie recommender (Naive Bayes)	Classification fundamentals, probability, and model fine-tuning
2	Ad click-through prediction	Trees, random forests, XGBoost, logistic regression at scale, feature importance
3	Stock-price regression	Regression modeling, residual diagnostics, regularization, time-aware evaluation
4	Face recognition + clustering	SVMs, multiclass classification, and unsupervised learning
5	Fashion image CNN	Convolutional networks, data augmentation, transfer learning in PyTorch
6	Sentiment + text generation (RNN)	Sequence modeling, LSTMs, and working with text data
7	GAN + CLIP image search	Generative modeling and modern multimodal AI
8	CAPSTONE (your choice)	End-to-end ownership: framing, building, evaluating, and communicating a real solution

Choosing your capstone (Day 25)

Pick something you can genuinely finish in three days and would enjoy talking about. Good capstones are specific, use a real dataset, and have a clear success metric. A few directions:

- A tabular prediction problem end-to-end with a deployed demo (e.g., predicting customer churn, loan default, or house prices) — shows you can ship.
- An image classifier for a domain you care about, using transfer learning — shows deep-learning competence.
- An NLP app: a fine-tuned sentiment/classification model or a small retrieval + LLM demo — shows you are current.
- A Kaggle competition entry with a documented approach and leaderboard score — shows you can compete.

The interview-prep track

Interview readiness is built in daily 1.5-hour doses, not crammed at the end. Each week the review block targets a theme so that by Week 4 you are polishing, not learning.

Week	Interview theme	Prepare to answer / do
1	Statistics & fundamentals	Bias/variance, sampling, CLT, p-values, confidence intervals, correlation vs causation, when a result is 'significant'
2	Classical ML & modeling	How each algorithm works, overfitting & regularization, cross-validation, metric choice, handling imbalanced data, feature engineering
3	Deep learning	Backprop, activation & loss functions, vanishing gradients, CNN vs RNN vs transformer, regularization (dropout, batchnorm), transfer learning
4	Applied & behavioral	ML system design, framing a problem, a coding drill a day, 'walk me through a project', and behavioral (STAR) stories

Daily interview habits

- **Explain-aloud:** end every day by teaching the day's core idea to an imaginary interviewer in 2–3 minutes.
- **A coding rep a day:** one short Python/pandas/NumPy or ML-implementation problem (e.g., code k-means or a train/test split from scratch).
- **A flashcard deck:** add 3–5 cards daily (Anki or paper) for definitions and 'why' questions; review the deck each morning.
- **Project talk-track:** for every project, write 4 sentences — problem, approach, result, what you'd do next. Interviewers love this.

Job-readiness checklist

Tick these off as you go. Being 'job-ready' is this list, not a feeling.

- ☐ GitHub profile with 8 clean, documented project repos
- ☐ One polished capstone with a write-up and a runnable demo
- ☐ A resume tuned to Applied ML / Data roles, with projects front and center
- ☐ LinkedIn updated: headline, skills, and featured projects
- ☐ A 2–3 minute talk-track for every project, rehearsed out loud
- ☐ A flashcard deck covering stats, classical ML, and deep learning
- ☐ Comfort implementing core algorithms from scratch (k-means, logistic regression, a small NN)

- ☐ At least one timed mock interview completed and reviewed
- ☐ A shortlist of 15–20 target companies/roles and a weekly application cadence
- ☐ Optional but strong: a Kaggle competition entry or a published Kaggle notebook

Staying on track

If you fall behind. You will have hard days — that is normal at this pace. Do not try to 'make up' by skipping the build blocks; understanding compounds. Instead, protect the core: prioritize the ML (spine) and BE (project) chapters, and treat the deepest PS sections as a second pass you can revisit in Week 4's review. It is better to deeply learn 80% than to skim 100%.

If a day feels too easy. Go deeper, not faster. Re-derive the math, break your code and fix it, or enter the relevant Kaggle competition for that technique. Depth is what interviews reward.

Weekly reset. Every Sunday (or your Day 7/14/21/28), spend 30 minutes reviewing your journal, re-running your flashcards, and adjusting the coming week. Momentum is the whole game.

You have a real, concrete path now. Show up for the 10 hours, ship something every day, and in 30 days you will be a different practitioner. Good luck, Dan.